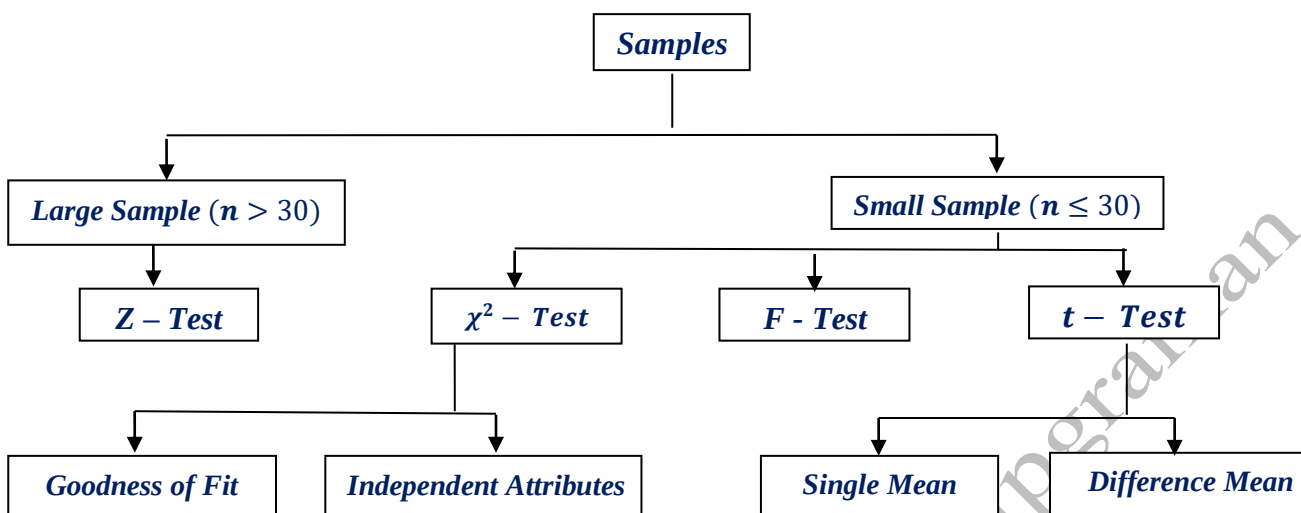


## TESTING OF HYPOTHESIS



## STATISTICS BASIC DEFINITIONS

### Population:

The word population or Universe in statistics is used to refer to any collection of individuals. The population may be finite or infinite. For example, we may have population of height, weight, ages, etc.

**Sample :** A finite subset of a population is called a sample.

**Sample size:** The number of elements in a sample is called the sample size.

### Sampling :

The process of selection of such samples is called sampling. For example, a housewife normally tests the cooked products to find if they are properly cooked & contain the proper quantity of salt.

**Parameters :** The statistical constants of the population such as the mean, the variance etc,

### Statistics :

The statistical concepts of the sample computed from the members or observation of the sample to estimate the parameters of the population from which the sample has been drawn are known as statistics. Population mean and variance are normally referred by  $\mu$  and  $\sigma^2$  while the sample mean and variance are referred by  $\bar{x}$  and  $s^2$ .

### Sampling Distribution :

If we draw a sample of size  $n$  from a given finite population of size  $N$  then the total number of possible sample is  $NC_n$ .  $NC_n = \frac{N!}{n!(N-n)!} = K$ .

**Standard Error (S.E.) :** The standard deviation of sampling distribution of a statistic is known as its standard error.

### Test of Significance :

A very important aspect of the sampling theory is the study of tests of significance which enable us to decide on the basis of the sample results if

- (i) The deviation between the observed sample statistic & the hypothetical parameter value is significant.
- (ii) The deviation between two sample statistics is significant.

### Null Hypothesis( $H_0$ ) :

There is no significant difference between the sample statistic and the corresponding population parameter or between two sample statistics is called Null Hypothesis.

**Alternate Hypothesis( $H_1$ ) :** A hypothesis that is different from the null hypothesis.

**Two Tailed :**  $H_0: \mu = \mu_0$ ,  $H_1: \mu \neq \mu_0$ ,

**One Tailed :**  $H_0: \mu = \mu_0$ ,  $H_1: \mu > \mu_0$  (**Right Tailed**),  $H_1: \mu < \mu_0$  (**Left Tailed**)

**Error in Sampling:** The main aim of the sampling theory is to draw a valid conclusion or a valid inference about the population parameters on the basis of the sample results.

**For example,** the mother at home tests the cooked products by taking and testing a small amount of cooked product. If this small amount of cooked product is good, we accept the lot to be good.

**Type I Error** : Reject  $H_0$  when it is true.

**Type II Error** : Accept  $H_0$  when it is wrong.

**Critical Region** : A region, corresponding to a statics  $t$ , in the sample space  $S$  which amounts to rejection of the null hypothesis  $H_0$  is called as critical region or region of rejection. The region of the sample space  $S$  which amounts to the acceptance  $H_0$  is called acceptance region.

**Level of Significance** : The probability  $\alpha$  that a random value of the statistic  $t$  belongs to the critical region is known as the level of significance. In other words, level of significance is the size of the Type I error. The levels of significance usually employed in testing of hypothesis are 5% and 1%.

### **PROCEDURE FOR TESTING OF HYPOTHESIS**

1. Set up the null hypothesis  $H_0$ .
2. Set up the alternative hypothesis  $H_1$ .  
This will enable us to decide whether we have to use a single tailed (right or left) test or two tailed test.
3. Choose the appropriate level of significance (either 5% or 1% level). This is to be decided before sample is drawn.
4. Compute the test statistics  $z = \frac{t - E(t)}{SE(t)}$  under the null hypothesis.
5. We compare the computed value of  $z$  in step (4) with the tabulated value  $z_\alpha$  at given level of significance  $\alpha$ .  
If the calculated value of  $Z$  is less than tabulated value  $z_\alpha$  then  $H_0$  accepted.  
If the calculated value of  $Z$  is greater than tabulated value  $z_\alpha$  then  $H_0$  rejected.

### **SMALL SAMPLES ( $n \leq 30$ )**

#### **Students 't' - Test - SINGLE MEAN**

**Standard deviation given directly:**  $t = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n-1}}\right)}$ , where  $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$ ,

$\bar{x}$  - sample mean,  $\mu$  - population mean,  $n$  - sample size,  $s^2$  - sample variance

**Standard deviation not given directly:**  $t = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n}}\right)}$ , where  $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$

$\bar{x}$  - sample mean,  $\mu$  - population mean,  $n$  - sample size,  $S^2$  - population variance  $\sigma^2$ , **Degree of freedom :  $n - 1$**

#### **DIFFERENCE MEAN**

$t = \frac{\bar{x} - \bar{y}}{\sqrt{S^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$ , where  $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$ ,  $\bar{y} = \frac{\sum_{j=1}^n y_j}{n}$ , **Degree of freedom :  $n_1 + n_2 - 2$**

**Standard deviation given directly:**  $S^2 = \frac{[n_1 s_1^2 + n_2 s_2^2]}{n_1 + n_2 - 2}$

**Standard deviation not given directly:**  $S^2 = \frac{\sum (x - \bar{x})^2 + \sum (y - \bar{y})^2}{n_1 + n_2 - 2}$ ,

#### **Confidence or Fiducial Limits for $\mu$ :**

95% Confidence limits for  $\mu$  :  $\bar{x} - t_{0.05} \left(\frac{s}{\sqrt{n}}\right) \leq \mu \leq \bar{x} + t_{0.05} \left(\frac{s}{\sqrt{n}}\right)$

99% Confidence limits for  $\mu$  :  $\bar{x} - t_{0.01} \left(\frac{s}{\sqrt{n}}\right) \leq \mu \leq \bar{x} + t_{0.01} \left(\frac{s}{\sqrt{n}}\right)$

#### **Application of t - distribution:**

The t - distribution has a wide number of applications in statistics, some of which are enumerated below

- (i) To test if the sample mean  $\bar{x}$  differs significantly from the hypothetical value  $\mu$  of the population mean.
- (ii) To test the significance of the difference between two sample means.
- (iii) To test the significance of an observed sample correlation coefficient and sample regression coefficient
- (iv) To test the significance of observed partial correlation coefficient.

#### **Assumption for students's t - test:**

- (i) The parent population from which the sample is drawn is normal.
- (ii) The sample observations are independent, that is, the sample is random.
- (iii) The population standard deviation  $\sigma$  is unknown.

### PROBLEMS

#### Students 't' – Test : Single Mean and Standard Deviation Given Directly

1. The mean weekly sales of soap bars in departmental stores was 146.3 bars per store. After an advertising campaign the mean weekly sales in 22 stores for a typical week increased to 153.7 and showed a S.D. of 17.2. Was the advertising campaign successful?

**Solution:**  $n = 22$ ,  $\bar{x} = 153.7$ ,  $\mu = 146.3$ ,  $s = 17.2$

**Null Hypothesis**  $H_0 : \mu = 146.3$ , i.e. The advertising campaign is not successful.

**Alternate Hypothesis**  $H_1 : \mu > 146.3$  (Right tailed)

$$t = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n-1}}\right)} = \frac{153.7 - 146.3}{\left(\frac{17.2}{\sqrt{22-1}}\right)} = 1.96, \text{ degree of freedom: } n - 1 = 21 \text{ at } 5\% \text{ level of significance} = 1.72.$$

Calculate value  $t >$  Tabulated  $t$ .  **$H_0$  is rejected.**

**Conclusion:** The advertising campaign was definitely successful in promoting sales.

2. A sample of 20 items has mean 42 units and S.D. 5 units. Test the hypothesis that it is a random sample from a normal population with mean 45 units.

**Solution:**  $n = 20$ ,  $\bar{x} = 42$ ,  $\mu = 45$ ,  $s = 5$

**Null Hypothesis**  $H_0 : \mu = 45$ , **Alternate Hypothesis**  $H_1 : \mu \neq 45$  (Two tailed)

$$t = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n-1}}\right)} = \frac{42 - 45}{\left(\frac{5}{\sqrt{20-1}}\right)} = 2.615, \text{ degree of freedom: } n - 1 = 19 \text{ at } 5\% \text{ level of significance} = 2.09.$$

Calculate value  $t >$  Tabulated  $t$ .  **$H_0$  is rejected.** The sample could not have come from this population.

3. The mean life time of a sample of 25 bulbs is found as 1550 hours with a S.D. of 120 hours. The company manufacturing the bulbs claims that the average life of their bulbs is 1600 hours. Is the claim acceptable at 5% level of significance?

**Solution:**  $n = 25$ ,  $\bar{x} = 1550$ ,  $\mu = 1600$ ,  $s = 120$

**Null Hypothesis**  $H_0 : \bar{x} = \mu$ , **Alternate Hypothesis**  $H_1 : \bar{x} < \mu$  (Left Tailed)

$$t = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n-1}}\right)} = \frac{1550 - 1600}{\left(\frac{120}{\sqrt{25-1}}\right)} = -2.04, \text{ degree of freedom: } n - 1 = 24 \text{ at } 5\% \text{ level of significance} = 1.71.$$

Calculate value  $t >$  Tabulated  $t$ .  **$H_0$  is rejected.** The claim of the company cannot be accepted at 5% LOS.

#### Students 't' – Test : Single Mean and Standard Deviation Not Given Directly

4. A random sample of 10 boys had the following I.Q.'s 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these data support the assumption of a population mean I.Q.'s of 100? Find a reasonable range in which most of the mean I.Q. values of samples of 10 boys lie.

**Solution:**  $n = 10$ ,  $\mu = 100$ ,  **$H_0 : \mu = 100$** ,  **$H_1 : \mu \neq 100$**  (Two tail)

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{972}{10} = 97.2, \quad S^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} = \frac{1833.6}{10-1} = 203.73$$

| $x$            | $x - \bar{x}$ | $(x - \bar{x})^2$                |
|----------------|---------------|----------------------------------|
| 70             | -27.2         | 739.84                           |
| 120            | 22.8          | 519.84                           |
| 110            | 12.8          | 163.84                           |
| 101            | 3.8           | 14.44                            |
| 88             | -9.2          | 84.64                            |
| 83             | -14.2         | 201.64                           |
| 95             | -2.2          | 4.84                             |
| 98             | 0.8           | 0.64                             |
| 107            | 9.8           | 96.04                            |
| 100            | 2.8           | 7.84                             |
| $\sum x = 972$ |               | $\sum (x - \bar{x})^2 = 1833.60$ |

$$t = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n}}\right)} = \frac{97.2 - 100}{\left(\frac{\sqrt{203.73}}{\sqrt{10}}\right)} = \frac{-2.8}{\left(\frac{\sqrt{203.73}}{10}\right)} = -0.62 \Rightarrow |t| = 0.62 ; \text{ d.f.} = n - 1 = 10 - 1 = 9 \text{ at } 5\% \text{ LOS} = 2.262.$$

Calculate value  $t <$  Tabulated  $t$ .  **$H_0$  is accepted.**

**Conclusion:** The data are consistent with the assumption for a mean I.Q. of 100 in population.

**95% Confidence Limits for  $\mu$ :**  $\bar{x} - t_{0.05} \left( \frac{s}{\sqrt{n}} \right) \leq \mu \leq \bar{x} + t_{0.05} \left( \frac{s}{\sqrt{n}} \right)$

$$97.2 - 2.262 \left( \sqrt{\frac{203.73}{10}} \right) \leq \mu \leq 97.2 + 2.262 \left( \sqrt{\frac{203.73}{10}} \right)$$

$86.99 \leq \mu \leq 107.41$ , Hence the required 95% confidence interval is [86.99, 107.41].

**Students 't' – Test : Difference Mean and Standard Deviation Given Directly**

**5. Sample of two types of electric light bulbs were tested for length of life and following data were obtained**

|               | Type I                         | Type II                        |
|---------------|--------------------------------|--------------------------------|
| Sample Number | $n_1 = 8$                      | $n_2 = 7$                      |
| Sample Means  | $\bar{x}_1 = 1234 \text{ hrs}$ | $\bar{x}_2 = 1036 \text{ hrs}$ |
| Sample S.D.   | $s_1 = 36 \text{ hrs}$         | $s_2 = 40 \text{ hrs}$         |

**Is the difference in the means sufficient to warrant that type I is superior to type II regarding length of life**

**Solution:** Null Hypothesis :  $H_0: \bar{x}_1 = \bar{x}_2$ , i.e. The two types I and II of electric bulbs are identical.

Alternate Hypothesis :  $H_1: \bar{x}_1 > \bar{x}_2$  (Right tail)

Standard deviation given directly:  $s^2 = \frac{[n_1 s_1^2 + n_2 s_2^2]}{n_1 + n_2 - 2} = \frac{[8(36)^2 + 7(40)^2]}{8 + 7 - 2} = 1659.08$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s^2 \left( \frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{1234 - 1036}{\sqrt{1659.08 \left( \frac{1}{8} + \frac{1}{7} \right)}} = 9.39, \text{ degree of freedom} = n_1 + n_2 - 2 = 3 \text{ at } 5\% = 1.77.$$

Calculate value  $t > \text{Tabulated } t$ .  **$H_0$  is rejected.** The type I is definitely superior to type II regarding length of life.

**Students 't' – Test : Difference Mean and Standard Deviation Not Given Directly**

**6. Below are given the gain in weights (in kgs) of pigs fed on two diets A and B.**

|        |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Diet A | 25 | 32 | 30 | 34 | 24 | 14 | 32 | 24 | 30 | 31 | 35 | 25 | -  | -  | -  |
| Diet B | 44 | 34 | 22 | 10 | 47 | 31 | 40 | 30 | 32 | 35 | 18 | 21 | 35 | 29 | 22 |

**Test if the two diets differ significantly as regards their effect on increase in weight.**

**Solution:** Null Hypothesis  $H_0: \bar{x}_1 = \bar{x}_2$ , i.e. There is no significant difference between the mean increase in weight due to diets A & B, Alternate Hypothesis :  $H_1: \bar{x}_1 \neq \bar{x}_2$  (Two tailed)

| $x$              | $x - \bar{x}$ | $(x - \bar{x})^2$             | $y$              | $y - \bar{y}$ | $(y - \bar{y})^2$              |
|------------------|---------------|-------------------------------|------------------|---------------|--------------------------------|
| 25               | -3            | 9                             | 44               | 14            | 196                            |
| 32               | 4             | 16                            | 34               | 4             | 16                             |
| 30               | 2             | 4                             | 22               | -8            | 64                             |
| 34               | 6             | 36                            | 10               | -20           | 400                            |
| 24               | -4            | 16                            | 47               | 17            | 289                            |
| 14               | -14           | 196                           | 31               | 1             | 1                              |
| 32               | 4             | 16                            | 40               | 10            | 100                            |
| 24               | -4            | 16                            | 30               | 0             | 0                              |
| 30               | 2             | 4                             | 32               | 2             | 4                              |
| 31               | 3             | 9                             | 35               | 5             | 25                             |
| 35               | 7             | 49                            | 18               | -12           | 144                            |
| 25               | -3            | 9                             | 21               | -9            | 81                             |
| $\Sigma x = 336$ |               | $\Sigma(x - \bar{x})^2 = 380$ | 35               | 5             | 25                             |
|                  |               |                               | 29               | -1            | 1                              |
|                  |               |                               | 22               | -8            | 64                             |
|                  |               |                               | $\Sigma y = 450$ |               | $\Sigma(y - \bar{y})^2 = 1410$ |

$$n_1 = 12, n_2 = 15, \bar{x} = \frac{\Sigma_{i=1}^n x_i}{n} = \frac{336}{12} = 28, \bar{y} = \frac{\Sigma_{j=1}^n y_j}{n} = \frac{450}{15} = 30$$

Standard deviation not given directly:  $s^2 = \frac{\Sigma(x - \bar{x})^2 + \Sigma(y - \bar{y})^2}{n_1 + n_2 - 2} = 71.6$

$$t = \frac{\bar{x} - \bar{y}}{\sqrt{s^2 \left( \frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{28 - 30}{\sqrt{71.6 \left( \frac{1}{12} + \frac{1}{15} \right)}} = -0.609 \Rightarrow |t| = 0.609, \text{ d.f.} = n_1 + n_2 - 2 = 25 \text{ at } 5\% \text{ LOS} = 2.06.$$

Calculate value  $t < \text{Tabulated } t$ .  **$H_0$  is accepted.**

**Conclusion:** The two diets do not differ significantly as regards their effect on increase in weight.

7. A group of five patients treated with medicine A weigh 42, 39, 48, 60 and 41 kg.; a second group of 7 patients from the same hospital treated with medicine B weigh 38, 42, 56, 64, 68, 69 and 62 kg. Do you agree with the claim that medicine B increases the weight significantly?

**Solution:** Null Hypothesis :  $H_0: \mu_1 = \mu_2$ , Alternate Hypothesis :  $H_1: \mu_1 < \mu_2$

| $x$              | $x - \bar{x}$ | $(x - \bar{x})^2$             | $y$              | $y - \bar{y}$ | $(y - \bar{y})^2$             |
|------------------|---------------|-------------------------------|------------------|---------------|-------------------------------|
| 42               | -4            | 16                            | 38               | -19           | 361                           |
| 39               | -7            | 49                            | 42               | -15           | 225                           |
| 48               | 2             | 4                             | 56               | -1            | 1                             |
| 60               | 14            | 196                           | 64               | 7             | 49                            |
| 41               | -5            | 25                            | 68               | 11            | 121                           |
| $\Sigma x = 230$ |               | $\Sigma(x - \bar{x})^2 = 290$ | 69               | 12            | 144                           |
|                  |               |                               | 62               | 5             | 25                            |
|                  |               |                               | $\Sigma y = 399$ |               | $\Sigma(y - \bar{y})^2 = 926$ |

$$n_1 = 5, n_2 = 7, \bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{230}{5} = 46, \bar{y} = \frac{\sum_{j=1}^n y_j}{n} = \frac{399}{7} = 57$$

**Standard deviation not given directly:**  $s^2 = \frac{\Sigma(x - \bar{x})^2 + \Sigma(y - \bar{y})^2}{n_1 + n_2 - 2} = \frac{290 + 926}{5 + 7 - 2} = \frac{1216}{10} = 121.6$

$$t = \frac{\bar{x} - \bar{y}}{\sqrt{s^2 \left( \frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{46 - 57}{\sqrt{121.6 \left( \frac{1}{5} + \frac{1}{7} \right)}} = \frac{46 - 57}{\sqrt{41.69}} = \frac{-11}{6.46} = -1.703 \Rightarrow |t| = 1.703, \text{ d.f.} = n_1 + n_2 - 2 = 10 \text{ at } 5\% = 1.81.$$

Calculate value  $t < \text{Tabulated } t$ .  **$H_0$  is accepted.** Medicines A and B do not differ significantly.

8. The table below represent the values of protein content from cow's milk and buffalo's milk at a certain level. Examine if these differences are significant.

|                       |             |             |             |             |             |             |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| <b>Cow's milk</b>     | <b>1.82</b> | <b>2.02</b> | <b>1.88</b> | <b>1.61</b> | <b>1.81</b> | <b>1.54</b> |
| <b>Buffalo's milk</b> | <b>2</b>    | <b>1.83</b> | <b>1.86</b> | <b>2.03</b> | <b>2.19</b> | <b>1.88</b> |

**Solution:** Null Hypothesis  $H_0: \bar{x}_1 = \bar{x}_2$ , Alternate Hypothesis :  $H_1: \bar{x}_1 \neq \bar{x}_2$  (Two tailed)

| $x_1$                | $x_1^2$                  | $x_2$                | $x_2^2$                  |
|----------------------|--------------------------|----------------------|--------------------------|
| 1.82                 | 3.3124                   | 2                    | 4.0040                   |
| 2.02                 | 4.0804                   | 1.83                 | 3.3489                   |
| 1.88                 | 3.5344                   | 1.86                 | 3.4596                   |
| 1.61                 | 2.5921                   | 2.03                 | 4.1209                   |
| 1.81                 | 3.2761                   | 2.19                 | 4.761                    |
| 1.54                 | 2.3716                   | 1.88                 | 3.5344                   |
| $\Sigma x_1 = 10.68$ | $\Sigma x_1^2 = 19.1670$ | $\Sigma x_2 = 11.79$ | $\Sigma x_2^2 = 23.2599$ |

$$n_1 = 6, n_2 = 6, \bar{x}_1 = \frac{\Sigma x_1}{n_1} = \frac{10.68}{6} = 1.78; \bar{x}_2 = \frac{\Sigma x_2}{n_2} = \frac{11.79}{6} = 1.965$$

**Sample S.D.:**  $s_1^2 = \frac{\Sigma x_1^2}{n_1} - (\bar{x}_1)^2 = \frac{19.167}{6} - (1.78)^2 = 0.0261; s_2^2 = \frac{\Sigma x_2^2}{n_2} - (\bar{x}_2)^2 = \frac{23.2599}{6} - (1.965)^2 = 0.0154$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\left( \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} \right) \left( \frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{1.78 - 1.965}{\sqrt{\left( \frac{6 \times 0.0261 + 6 \times 0.0154}{6 + 6 - 2} \right) \left( \frac{1}{6} + \frac{1}{6} \right)}} = -2.03 \Rightarrow |t| = 2.03, \text{ d.f.} = n_1 + n_2 - 2 = 10 \text{ at } 5\% = 2.26.$$

Calculate value  $t < \text{Tabulated } t$ .  **$H_0$  is accepted.**

**Conclusion:** The difference between the mean protein values of the two varieties of milk is not significant at 5%.

### F – Test

$$F = \frac{S_X^2}{S_Y^2}, \text{ if } S_X^2 > S_Y^2 \text{ (or)} = \frac{S_Y^2}{S_X^2}, \text{ if } S_X^2 < S_Y^2,$$

$$S_X^2 = \frac{\sum(x-\bar{x})^2}{n_1-1}, S_Y^2 = \frac{\sum(y-\bar{y})^2}{n_2-1}, S_1^2 = \frac{n_1 s_1^2}{n_1-1}, S_2^2 = \frac{n_2 s_2^2}{n_2-1},$$

degree of freedom : ( $\nu_1 = n_1 - 1, \nu_2 = n_2 - 1$ ).

**Applications:** (i) To test whether two independent samples have been drawn from the normal populations with the same variance  $\sigma^2$  (ii) To test whether two independent estimates of the population variance are homogeneous or not.

### PROBLEMS

9. If one sample of 8 observations the sum of the squares of deviations of the sample values from the sample mean was 84.4 and in the other sample of 10 observations it was 102.6. Test whether this difference is significant at 5%.

**Solution:**  $n_1 = 8, n_2 = 10, \sum(x - \bar{x})^2 = 84.4, \sum(y - \bar{y})^2 = 102.6$

Null Hypothesis :  $H_0: \sigma_X^2 = \sigma_Y^2$ , Alternate Hypothesis :  $H_1: \sigma_X^2 \neq \sigma_Y^2$

$$S_X^2 = \frac{\sum(x-\bar{x})^2}{n_1-1} = \frac{84.4}{8-1} = 12.057, S_Y^2 = \frac{\sum(y-\bar{y})^2}{n_2-1} = \frac{102.6}{10-1} = 11.4, F = \frac{S_X^2}{S_Y^2} = \frac{12.057}{11.4} = 1.057,$$

d.f. = ( $n_1 - 1, n_2 - 1$ ) = (7, 9) at 5% LOS = 3.29. Calculate value  $F < \text{Tabulated } F$ .  **$H_0$  is accepted.**

10. Two random samples of 11 and 9 items show the sample standard deviations of their weights as 0.8 and 0.5 respectively. Assuming that the weight distributions are normal, test the hypothesis that the true variances are equal, against the alternative hypothesis that they are not.

**Solution:**  $n_1 = 11, n_2 = 9, s_1^2 = 0.8, s_2^2 = 0.5$ , Null Hypothesis:  $H_0: \sigma_X^2 = \sigma_Y^2$ , Alternate Hypothesis :  $H_1: \sigma_X^2 \neq \sigma_Y^2$

$$S_1^2 = \frac{n_1 s_1^2}{n_1-1} = \frac{11(0.8)^2}{11-1} = 0.704, S_2^2 = \frac{n_2 s_2^2}{n_2-1} = \frac{9(0.5)^2}{9-1} = 0.28125, F = \frac{S_1^2}{S_2^2} = \frac{0.704}{0.28125} = 2.503,$$

d.f. = ( $n_1 - 1, n_2 - 1$ ) = (10, 8) at 5% LOS = 3.34. Calculate value  $F < \text{Tabulated } F$ .  **$H_0$  is accepted.**

11. The time taken by workers in performing a job by Method I and Method II is given below

|           |    |    |    |    |    |    |    |
|-----------|----|----|----|----|----|----|----|
| Method I  | 20 | 16 | 26 | 27 | 23 | 22 | -  |
| Method II | 27 | 33 | 42 | 35 | 32 | 34 | 38 |

Do the data show that the variances of time distribution form population from which these samples are drawn do not differ significantly?

**Solution:**  $n_1 = 6, n_2 = 7, \bar{x} = \frac{\sum_{i=1}^n x_i}{n_1} = \frac{134}{6} = 22.3, \bar{y} = \frac{\sum_{j=1}^n y_j}{n_2} = \frac{241}{7} = 34.4$

| $x$            | $x - \bar{x}$ | $(x - \bar{x})^2$             | $y$            | $y - \bar{y}$ | $(y - \bar{y})^2$              |
|----------------|---------------|-------------------------------|----------------|---------------|--------------------------------|
| 20             | -2.3          | 5.29                          | 27             | -7.4          | 54.76                          |
| 16             | -6.3          | 39.69                         | 33             | -1.4          | 1.96                           |
| 26             | 3.7           | 13.69                         | 42             | 7.6           | 57.76                          |
| 27             | 4.7           | 22.09                         | 35             | 0.6           | 0.36                           |
| 23             | 0.7           | 0.49                          | 32             | -2.4          | 5.76                           |
| 22             | -0.3          | 0.09                          | 34             | -0.4          | 0.16                           |
| $\sum x = 134$ |               | $\sum(x - \bar{x})^2 = 81.34$ | 38             | 3.6           | 12.96                          |
|                |               |                               | $\sum y = 241$ |               | $\sum(y - \bar{y})^2 = 132.72$ |

Null Hypothesis :  $H_0: \sigma_X^2 = \sigma_Y^2$ , Alternate Hypothesis :  $H_1: \sigma_X^2 \neq \sigma_Y^2$

$$S_X^2 = \frac{\sum(x-\bar{x})^2}{n_1-1} = \frac{81.34}{6-1} = 16.268, S_Y^2 = \frac{\sum(y-\bar{y})^2}{n_2-1} = \frac{132.72}{7-1} = 22.29, F = \frac{S_Y^2}{S_X^2} = \frac{22.29}{16.268} = 1.37,$$

d.f. = ( $n_2 - 1, n_1 - 1$ ) = (6, 5) at 5% LOS = 4.95. Calculate value  $F < \text{Tabulated } F$ .  **$H_0$  is accepted.**

12. Two horses A and B were tested according to the time (in seconds) to run a particular track with the following

|         |    |    |    |    |    |    |    |
|---------|----|----|----|----|----|----|----|
| Horse A | 28 | 30 | 32 | 33 | 33 | 29 | 34 |
| Horse B | 29 | 30 | 30 | 24 | 27 | 29 | -  |

Test whether the two horses have the same running capacity.

**Solution:**  $n_1 = 7, n_2 = 6, \bar{x} = \frac{\sum_{i=1}^n x_i}{n_1} = \frac{219}{7} = 31.28, \bar{y} = \frac{\sum_{j=1}^n y_j}{n_2} = \frac{169}{6} = 28.2$



| $x$              | $x - \bar{x}$ | $(x - \bar{x})^2$              | $y$              | $y - \bar{y}$ | $(y - \bar{y})^2$             |
|------------------|---------------|--------------------------------|------------------|---------------|-------------------------------|
| 28               | -3.28         | 10.75                          | 29               | 0.8           | 0.64                          |
| 30               | -1.28         | 1.64                           | 30               | 1.8           | 3.24                          |
| 32               | 0.72          | 0.52                           | 30               | 1.8           | 3.24                          |
| 33               | 1.72          | 2.96                           | 24               | -4.2          | 17.64                         |
| 33               | 1.72          | 2.96                           | 27               | -1.2          | 1.44                          |
| 29               | -2.28         | 5.20                           | 29               | 0.8           | 0.64                          |
| 34               | 2.72          | 7.40                           | -                | -             | -                             |
| $\Sigma x = 219$ |               | $\Sigma(x - \bar{x})^2 = 21.2$ | $\Sigma y = 169$ |               | $\Sigma(y - \bar{y})^2 = 108$ |

#### F - Test

Null Hypothesis :  $H_0: \sigma_X^2 = \sigma_Y^2$ , Alternate Hypothesis :  $H_1: \sigma_X^2 \neq \sigma_Y^2$

$$S_X^2 = \frac{\Sigma(x - \bar{x})^2}{n_1 - 1} = \frac{21.43}{7 - 1} = 5.238, S_Y^2 = \frac{\Sigma(y - \bar{y})^2}{n_2 - 1} = \frac{26.84}{6 - 1} = 5.368, F = \frac{S_Y^2}{S_X^2} = \frac{5.368}{5.238} = 1.02$$

degree of freedom =  $(n_2 - 1, n_1 - 1) = (5, 6)$  at 5% LOS = 4.39.

Calculate value  $F < \text{Tabulated } F$ .  **$H_0$  is accepted.** **Conclusion:** The two horses have the same running capacity.

#### t - test and F - Test

**13. The nicotine contents in milligrams in two samples of tobacco were found to be as follows**

|                 |           |           |           |           |           |           |
|-----------------|-----------|-----------|-----------|-----------|-----------|-----------|
| <b>Sample A</b> | <b>24</b> | <b>27</b> | <b>26</b> | <b>21</b> | <b>25</b> | <b>-</b>  |
| <b>Sample B</b> | <b>27</b> | <b>30</b> | <b>28</b> | <b>31</b> | <b>22</b> | <b>36</b> |

**Can it be said that two samples come from normal populations.**

**Solution:**  $n_1 = 5, n_2 = 6, \bar{x} = \frac{\Sigma_{i=1}^n x_i}{n_1} = \frac{123}{5} = 24.6, \bar{y} = \frac{\Sigma_{j=1}^n y_j}{n_2} = \frac{174}{6} = 29$

| $x$              | $x - \bar{x}$ | $(x - \bar{x})^2$              | $y$              | $y - \bar{y}$ | $(y - \bar{y})^2$             |
|------------------|---------------|--------------------------------|------------------|---------------|-------------------------------|
| 24               | -0.6          | 0.36                           | 27               | -2            | 4                             |
| 27               | 2.4           | 5.76                           | 30               | 1             | 1                             |
| 26               | 1.4           | 1.96                           | 28               | -1            | 1                             |
| 21               | -3.6          | 12.96                          | 31               | 2             | 4                             |
| 25               | 0.4           | 0.16                           | 22               | -7            | 49                            |
| $\Sigma x = 123$ |               | $\Sigma(x - \bar{x})^2 = 21.2$ | 36               | 7             | 49                            |
|                  |               |                                | $\Sigma y = 174$ |               | $\Sigma(y - \bar{y})^2 = 108$ |

#### Student's t test

Null Hypothesis :  $H_0: \bar{x}_1 = \bar{x}_2$ , Alternate Hypothesis :  $H_1: \bar{x}_1 \neq \bar{x}_2$  (Two tailed)

$$S^2 = \frac{\Sigma(x - \bar{x})^2 + \Sigma(y - \bar{y})^2}{n_1 + n_2 - 2} = \frac{21.2 + 108}{5 + 6 - 2} = 14.35, t = \frac{\bar{x} - \bar{y}}{\sqrt{S^2 \left( \frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{24.6 - 29}{\sqrt{14.35 \left( \frac{1}{5} + \frac{1}{6} \right)}} = -1.92 \Rightarrow |t| = 1.92$$

degree of freedom =  $n_1 + n_2 - 2 = 9$  at 5% LOS for two tailed test = 2.26.

Calculate value  $t < \text{Tabulated } t$ .  **$H_0$  is accepted.**

#### F - Test

Null Hypothesis :  $H_0: \sigma_X^2 = \sigma_Y^2$ , Alternate Hypothesis :  $H_1: \sigma_X^2 \neq \sigma_Y^2$

$$S_X^2 = \frac{\Sigma(x - \bar{x})^2}{n_1 - 1} = \frac{21.2}{5 - 1} = 5.3, S_Y^2 = \frac{\Sigma(y - \bar{y})^2}{n_2 - 1} = \frac{108}{6 - 1} = 21.6, F = \frac{S_Y^2}{S_X^2} = \frac{21.6}{5.3} = 4.07$$

degree of freedom =  $(n_2 - 1, n_1 - 1) = (5, 4)$  at 5% LOS = 6.26.

Calculate value  $F < \text{Tabulated } F$ .  **$H_0$  is accepted.**

**Conclusion:** We conclude that the two samples come from same normal population.

### $\chi^2$ – test (Chi square Test)

The  $\chi^2$  distribution function is one of the most extensively used distribution functions in statistics.

#### Application (or uses) of $\chi^2$ distribution

1. To test the goodness of fit.
2. To test the independence of attributes.
3. To test if the hypothetical value of the population variance is  $\sigma^2$ .
4. To test the homogeneity of independent estimates of the population variance.
5. To test the homogeneity of independent estimates of the population correlation coefficient.

#### Condition for validity of $\chi^2$ – test

1. The sample observations should be independent.
2. Constraints on the cell frequencies, if any, should be linear, e.g.,  $\sum O_i = \sum E_i$
3. N, the total frequency should be reasonably large, say, greater than 50.
4. No theoretical cell frequency should be less than 5. (The chi square distribution is frequency is less than t). If any theoretical cell frequency is less than 5, then for the application of  $\chi^2$  test, it is pooled with the preceding or succeeding frequency so that the pooled frequency is more than 5 and finally adjust for the degree of freedom lost in pooling.

#### $\chi^2$ – test of Goodness of fit

This is a powerful test for testing the significance of the discrepancy between theory and experiment – discovered by Karl Pearson in 1900. It helps us to find if the deviation of the experiment from theory is just by chance or it is due to the inadequacy of the theory to fit the observed data.

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}, \text{ where } O_i - \text{set of observed frequencies, } E_i - \text{Set of expected frequencies. d.f.} = n - 1$$

1. The following table gives the number of aircraft accidents that occurred during the various days of the week. Test whether the accidents are uniformly distributed over the week.

| Days             | Mon | Tue | Wed | Thu | Fri | Sat |
|------------------|-----|-----|-----|-----|-----|-----|
| No. of Accidents | 14  | 18  | 12  | 11  | 15  | 14  |

**Solution:**  $H_0$  : Accidents occur uniformly over the week. The expected no. of accidents on any day =  $\frac{84}{6} = 14$ .

| O  | E  | O – E | (O – E) <sup>2</sup> | $\frac{(O-E)^2}{E}$ |
|----|----|-------|----------------------|---------------------|
| 14 | 14 | 0     | 0                    | 0                   |
| 18 | 14 | 4     | 16                   | 1.143               |
| 12 | 14 | -2    | 4                    | 0.286               |
| 11 | 14 | -3    | 9                    | 0.643               |
| 15 | 14 | 1     | 1                    | 0.071               |
| 14 | 14 | 0     | 0                    | 0                   |
|    |    |       |                      | <b>2.143</b>        |

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} = 2.143, \text{ d.f.} = n - 1 = 5 \text{ at } 5\% \text{ LOS} = 11.07, \text{ Calculated } \chi^2 < \text{tabulated } \chi^2,$$

**$H_0$  is accepted.** The accidents are uniformly distributed over the week.



2. The theory predicts the proportion of beans, in the four groups A, B, C, and D should be 9 : 3 : 3 : 1. In an experiment with 1600 beans the numbers in the four groups were 882, 313, 287, and 118. Does the experimental result support the theory.

**Solution:**  $H_0$  : The experimental result support the theory.

| $O$ | $E$                              | $O - E$ | $(O - E)^2$ | $\frac{(O-E)^2}{E}$ |
|-----|----------------------------------|---------|-------------|---------------------|
| 882 | $\frac{9}{16} \times 1600 = 900$ | -18     | 324         | 0.36                |
| 313 | $\frac{3}{16} \times 1600 = 300$ | 13      | 169         | 0.563               |
| 287 | $\frac{3}{16} \times 1600 = 300$ | -13     | 169         | 0.563               |
| 118 | $\frac{1}{16} \times 1600 = 100$ | 18      | 324         | 3.24                |
|     |                                  |         |             | <b>4.726</b>        |

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} = 4.726, \text{ d.f.} = n - 1 = 4 - 1 = 3 \text{ at } 5\% \text{ LOS} = 7.81$$

Calculated  $\chi^2 < \text{tabulated } \chi^2$ ,  $H_0$  is accepted. The experimental results support the theory.

### $\chi^2$ – test of Independence of Attributes

Literally, an attribute means a quality or characteristic. E.g.: drinking, smoking, blindness, honesty, etc.

|           |           |           |
|-----------|-----------|-----------|
| $a$       | $b$       | $(a + b)$ |
| $c$       | $d$       | $(c + d)$ |
| $(a + c)$ | $(b + d)$ | $N$       |

|                                   |                                   |           |
|-----------------------------------|-----------------------------------|-----------|
| $E(a) = \frac{(a + c)(a + b)}{N}$ | $E(b) = \frac{(b + d)(a + b)}{N}$ | $(a + b)$ |
| $E(c) = \frac{(a + c)(c + d)}{N}$ | $E(d) = \frac{(b + d)(c + d)}{N}$ | $(c + d)$ |
| $(a + c)$                         | $(b + d)$                         | $N$       |

Degree of freedom =  $(r - 1)(c - 1)$ , where  $r$  – number of rows,  $c$  – number of columns.

3. On the basis of information given below about the treatment of 200 patients suffering from a disease, state whether the new treatment is comparatively superior to the conventional treatment.

|                     | <b>Favourable</b> | <b>Not Favourable</b> | <b>Total</b> |
|---------------------|-------------------|-----------------------|--------------|
| <b>New</b>          | <b>60</b>         | <b>30</b>             | <b>90</b>    |
| <b>Conventional</b> | <b>40</b>         | <b>70</b>             | <b>110</b>   |
|                     | <b>100</b>        | <b>100</b>            | <b>200</b>   |

**Solution:**  $H_0$  : New and conventional treatment are independent.

|                                           |                                           |            |
|-------------------------------------------|-------------------------------------------|------------|
| $E(60) = \frac{90 \times 100}{200} = 45$  | $E(30) = \frac{90 \times 100}{200} = 45$  | <b>90</b>  |
| $E(40) = \frac{110 \times 100}{200} = 55$ | $E(70) = \frac{110 \times 100}{200} = 55$ | <b>110</b> |
| <b>100</b>                                | <b>100</b>                                | <b>200</b> |

| $O$ | $E$ | $O - E$ | $(O - E)^2$ | $\frac{(O-E)^2}{E}$ |
|-----|-----|---------|-------------|---------------------|
| 60  | 45  | 15      | 225         | 5                   |
| 30  | 45  | -15     | 225         | 5                   |
| 40  | 55  | -15     | 225         | 4.09                |
| 70  | 55  | 15      | 225         | 4.09                |
|     |     |         |             | <b>18.18</b>        |

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} = 18.18$$

d.f. =  $(r - 1)(c - 1) = (2 - 1)(2 - 1) = 1$  at 5% LOS = 3.841.

Calculated  $\chi^2 >$  tabulated  $\chi^2$

**$H_0$  is rejected.**

New and conventional treatment are not independent.

### **CORRELATION**

**Correlation:** If the change in one variable affects a change in the other variable, the variables are said to be correlated.

#### **Types of Correlation**

1. **Positive Correlation :** If increase in one results in a corresponding increase in the other, correlation is said to be direct or positive. **Example :** (i) The heights and weights of a group of persons. (ii) The income and expenditure.
2. **Negative Correlation :** If increase in one results in a corresponding decrease in the other, correlation is said to be diverse or negative. **Example :** (i) Price and demand of a commodity (ii) The volume and pressure of a perfect gas.
3. **Simple Correlation :** When only two variables are studied it is a problem of simple correlation.
4. **Partial and Multiple Correlation:** When three or more variables are studied it is a problem of either multiple or partial correlation.
5. **Linear Correlation:** If the amount of change in any variable tends to bear a constant ratio to the amount of change in the other variable, then the correlation is said to be linear.
6. **Non-Linear Correlation:** If the amount of change in one variable does not bear a constant ratio to the amount of change in the other variable, then the correlation is said to be non-linear.

#### **Properties of Correlation Coefficient**

- (i) The coefficient of correlation lies between -1 and 1 or  $|r| \leq 1$ .

If  $r = 1$  then there is a perfect positive correlation

If  $r = -1$  then there is a perfect negative correlation

If  $r = 0$  then the variables are uncorrelated.

- (ii) The coefficient of correlation is independent of change of scale and origin of the variables  $X$  and  $Y$ .

#### **Karl Pearson's Coefficient of Correlation**

Correlation coefficient between two random variable  $X$  and  $Y$ , usually denoted by  $r(X, Y)$  or simply  $r_{XY}$ , is a numerical measure of linear relationship between them and is defined as

$$r(X, Y) = r_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\frac{1}{n} \sum XY - \bar{X} \bar{Y}}{\sqrt{\left(\frac{1}{n} \sum X^2 - \bar{X}^2\right) \left(\frac{1}{n} \sum Y^2 - \bar{Y}^2\right)}}$$

## PROBLEMS IN CORRELATION

1. Calculate the correlation coefficient for the following heights (in inches) of fathers (X) and their sons (Y):

|          |    |    |    |    |    |    |    |    |
|----------|----|----|----|----|----|----|----|----|
| <b>X</b> | 65 | 66 | 67 | 67 | 68 | 69 | 70 | 72 |
| <b>Y</b> | 67 | 68 | 65 | 68 | 72 | 72 | 69 | 71 |

**Solution:**

### Method - I

| <b>X</b>        | <b>Y</b>        | <b>XY</b>          | <b>X<sup>2</sup></b>          | <b>Y<sup>2</sup></b>          |
|-----------------|-----------------|--------------------|-------------------------------|-------------------------------|
| 65              | 67              | 4355               | 4225                          | 4489                          |
| 66              | 68              | 4488               | 4356                          | 4624                          |
| 67              | 65              | 4355               | 4489                          | 4225                          |
| 67              | 68              | 4556               | 4489                          | 4624                          |
| 68              | 72              | 4896               | 4624                          | 5184                          |
| 69              | 72              | 4968               | 4761                          | 5184                          |
| 70              | 69              | 4830               | 4900                          | 4761                          |
| 72              | 71              | 5112               | 5184                          | 5041                          |
| <b>ΣX = 544</b> | <b>ΣY = 552</b> | <b>ΣXY = 37560</b> | <b>ΣX<sup>2</sup> = 37028</b> | <b>ΣY<sup>2</sup> = 38132</b> |

$$\bar{X} = \frac{\Sigma X}{n} = \frac{544}{8} = 68, \quad \bar{Y} = \frac{\Sigma Y}{n} = \frac{552}{8} = 69$$

$$r(X, Y) = r_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\frac{1}{n} \Sigma XY - \bar{X} \bar{Y}}{\sqrt{\left(\frac{1}{n} \Sigma X^2 - \bar{X}^2\right) \left(\frac{1}{n} \Sigma Y^2 - \bar{Y}^2\right)}} = \frac{\frac{1}{8} \times 37560 - 68 \times 69}{\sqrt{\left(\frac{1}{8} \times 37028 - 68^2\right) \left(\frac{1}{8} \times 38132 - 69^2\right)}} = 0.603$$

### Method - II

| <b>X</b> | <b>Y</b> | <b>U = X - 68</b> | <b>V = Y - 69</b> | <b>UV</b>       | <b>U<sup>2</sup></b>       | <b>V<sup>2</sup></b>       |
|----------|----------|-------------------|-------------------|-----------------|----------------------------|----------------------------|
| 65       | 67       | -3                | -2                | 6               | 9                          | 4                          |
| 66       | 68       | -2                | -1                | 2               | 4                          | 1                          |
| 67       | 65       | -1                | -4                | 4               | 1                          | 16                         |
| 67       | 68       | -1                | -1                | 1               | 1                          | 1                          |
| 68       | 72       | 0                 | 3                 | 0               | 0                          | 9                          |
| 69       | 72       | 1                 | 3                 | 3               | 1                          | 9                          |
| 70       | 69       | 2                 | 0                 | 0               | 4                          | 0                          |
| 72       | 71       | 4                 | 2                 | 8               | 16                         | 4                          |
|          |          | <b>ΣU = 0</b>     | <b>ΣV = 0</b>     | <b>ΣUV = 24</b> | <b>ΣU<sup>2</sup> = 36</b> | <b>ΣV<sup>2</sup> = 44</b> |

$$\bar{U} = \frac{\Sigma U}{n} = \frac{0}{8} = 0, \quad \bar{V} = \frac{\Sigma V}{n} = \frac{0}{8} = 0$$

$$r(X, Y) = r(U, V) = r_{UV} = \frac{\text{Cov}(U, V)}{\sigma_U \sigma_V} = \frac{\frac{1}{n} \Sigma UV - \bar{U} \bar{V}}{\sqrt{\left(\frac{1}{n} \Sigma U^2 - \bar{U}^2\right) \left(\frac{1}{n} \Sigma V^2 - \bar{V}^2\right)}} = \frac{\frac{1}{8} \times 24 - 0}{\sqrt{\left(\frac{1}{8} \times 36 - 0\right) \left(\frac{1}{8} \times 44 - 0\right)}} = 0.603$$

2. Find the coefficient of correlation between industrial production and export using the following data

|                   |    |    |    |    |    |    |    |
|-------------------|----|----|----|----|----|----|----|
| <b>Production</b> | 55 | 56 | 58 | 59 | 60 | 60 | 62 |
| <b>Export</b>     | 35 | 38 | 37 | 39 | 44 | 43 | 44 |

**Solution:**  $\bar{U} = \frac{\Sigma U}{n} = \frac{-10}{7} = -1.43, \quad \bar{V} = \frac{\Sigma V}{n} = \frac{15}{7} = 2.14$

| <b>X</b> | <b>Y</b> | <b>U = X - 60</b> | <b>V = Y - 38</b> | <b>UV</b>       | <b>U<sup>2</sup></b>       | <b>V<sup>2</sup></b>        |
|----------|----------|-------------------|-------------------|-----------------|----------------------------|-----------------------------|
| 55       | 35       | -5                | -3                | 15              | 25                         | 9                           |
| 56       | 38       | -4                | 0                 | 0               | 16                         | 0                           |
| 58       | 37       | -2                | 0                 | 0               | 4                          | 0                           |
| 59       | 39       | -1                | 1                 | -1              | 1                          | 1                           |
| 60       | 44       | 0                 | 6                 | 0               | 0                          | 36                          |
| 60       | 43       | 0                 | 5                 | 0               | 0                          | 25                          |
| 62       | 44       | 2                 | 6                 | 12              | 4                          | 36                          |
|          |          | <b>ΣU = -10</b>   | <b>ΣV = 15</b>    | <b>ΣUV = 26</b> | <b>ΣU<sup>2</sup> = 50</b> | <b>ΣV<sup>2</sup> = 107</b> |

$$r(X, Y) = r(U, V) = r_{UV} = \frac{\text{Cov}(U, V)}{\sigma_U \sigma_V} = \frac{\frac{1}{n} \Sigma UV - \bar{U} \bar{V}}{\sqrt{\left(\frac{1}{n} \Sigma U^2 - \bar{U}^2\right) \left(\frac{1}{n} \Sigma V^2 - \bar{V}^2\right)}} = \frac{\frac{1}{7} \times 26 + 1.43}{\sqrt{\left(\frac{1}{7} \times 50 - 2.045\right) \left(\frac{1}{7} \times 107 - 4.58\right)}} = 0.92$$

## REGRESSION

Regression is the measure of the average relationship between the two or more variables in terms of original unit of data. For example, If the sales and advertising are correlated we can find out the expected amount of sales for a given advertising expenditure or the amount needed for attaining the given amount of sales.

### **Lines of Regression**

Equation of line of regression of Y on X :  $y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$

Equation of line of regression of X on Y :  $x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$

### **Regression coefficients**

$$b_{xy} = r \frac{\sigma_x}{\sigma_y} = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum y^2 - (\sum y)^2}, \quad b_{yx} = r \frac{\sigma_y}{\sigma_x} = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

### **Correlation coefficients**

$$r = \pm \sqrt{b_{xy} b_{yx}}$$

## PROBLEMS IN REGRESSION

### **1. Marks obtained by 10 students in Mathematics (x) and Statistics (y) are given below**

|   |    |    |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|----|----|
| x | 60 | 34 | 40 | 50 | 45 | 40 | 22 | 43 | 42 | 64 |
| y | 75 | 32 | 33 | 40 | 45 | 33 | 12 | 30 | 34 | 51 |

**Find the two regression lines . Also find y when x = 55**

**Solution:**  $\bar{x} = \frac{\sum x}{n} = \frac{440}{10} = 44, \quad \bar{y} = \frac{\sum y}{n} = \frac{385}{10} = 38.5$

| x              | y              | xy                | x <sup>2</sup>     | y <sup>2</sup>     |
|----------------|----------------|-------------------|--------------------|--------------------|
| 60             | 75             | 4500              | 3600               | 5625               |
| 34             | 32             | 1088              | 1156               | 1024               |
| 40             | 33             | 1320              | 1600               | 1089               |
| 50             | 40             | 2000              | 2500               | 1600               |
| 45             | 45             | 2025              | 2025               | 2025               |
| 40             | 33             | 1320              | 1600               | 1089               |
| 22             | 12             | 264               | 484                | 144                |
| 43             | 30             | 1290              | 1849               | 900                |
| 42             | 34             | 1428              | 1764               | 1156               |
| 64             | 51             | 3264              | 4096               | 2601               |
| $\sum x = 440$ | $\sum y = 385$ | $\sum xy = 18499$ | $\sum x^2 = 20674$ | $\sum y^2 = 17253$ |

$$b_{xy} = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum y^2 - (\sum y)^2} = \frac{(10 \times 18499) - (440 \times 385)}{(10 \times 17253) - (385)^2} = 0.6414$$

$$b_{yx} = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2} = \frac{(10 \times 18499) - (440 \times 385)}{(10 \times 20674) - (440)^2} = 1.1865$$

### **Regression line of Y on X**

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x}) \Rightarrow y - 38.5 = 1.1865(x - 44)$$

$$y = 1.1865x - 13.706 \quad \text{Where } x = 55, \quad y = (1.1865 \times 55) - 13.706 = 51.55$$

### **Regression line of X on Y**

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y}) \Rightarrow x - 44 = 0.6414(y - 38.5) \Rightarrow x = 0.6414y + 19.306$$

2. Obtain the equation of the regression lines from the following data. Hence find the coefficient of correlation between X and Y. Also estimate the value of Y when X = 38 and X when Y = 18.

|   |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|
| x | 22 | 26 | 29 | 30 | 31 | 31 | 34 | 35 |
| y | 20 | 20 | 21 | 29 | 27 | 24 | 27 | 31 |

**Solution:**

| x  | y  | u = x - 29     | v = y - 27       | uv               | u <sup>2</sup>     | v <sup>2</sup>     |
|----|----|----------------|------------------|------------------|--------------------|--------------------|
| 22 | 20 | -7             | -7               | 49               | 49                 | 49                 |
| 26 | 20 | -3             | -7               | 21               | 9                  | 49                 |
| 29 | 21 | 0              | -6               | 0                | 0                  | 36                 |
| 30 | 29 | 1              | 2                | 2                | 1                  | 4                  |
| 31 | 27 | 2              | 0                | 0                | 4                  | 0                  |
| 31 | 24 | 2              | -3               | -6               | 4                  | 9                  |
| 34 | 27 | 5              | 0                | 0                | 25                 | 0                  |
| 35 | 31 | 6              | 4                | 24               | 36                 | 16                 |
|    |    | $\Sigma u = 6$ | $\Sigma v = -17$ | $\Sigma uv = 90$ | $\Sigma u^2 = 128$ | $\Sigma v^2 = 163$ |

$$\bar{u} = \frac{\Sigma u}{n} = \frac{6}{8} = 0.75, \quad \bar{v} = \frac{\Sigma v}{n} = \frac{-17}{8} = -2.125$$

$$u = x - 29 \Rightarrow x = u + 29, \quad \bar{x} = 29 + \bar{u} = 29 + 0.75 = 30$$

$$v = y - 27 \Rightarrow y = v + 27, \quad \bar{y} = 27 + \bar{v} = 27 - 2.125 = 25$$

$$b_{xy} = b_{uv} = \frac{n \Sigma uv - (\Sigma u)(\Sigma v)}{n \Sigma v^2 - (\Sigma v)^2} = \frac{(8 \times 90) - (6 \times -17)}{(8 \times 163) - (-17)^2} = \frac{822}{1015} = 0.81$$

$$b_{yx} = b_{vu} = \frac{n \Sigma uv - (\Sigma u)(\Sigma v)}{n \Sigma u^2 - (\Sigma u)^2} = \frac{(8 \times 90) - (6 \times -17)}{(8 \times 128) - (6)^2} = \frac{822}{988} = 0.83$$

$$\text{Correlation coefficients : } r = \pm \sqrt{b_{xy} b_{yx}} = \pm \sqrt{0.81 \times 0.83} = 0.82$$

**Regression line of Y on X**

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x}) \Rightarrow y - 25 = 0.83 (x - 30) \Rightarrow y = 0.83 x - 0.1 \text{ When } x = 38, y = 31.44$$

**Regression line of X on Y**

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y}) \Rightarrow x - 30 = 0.81 (y - 25) \Rightarrow x = 0.81 y + 9.75 \text{ When } y = 18, x = 24.33$$

**DESIGN OF EXPERIMENTS**

There are situations which lead us to tests of hypotheses that involve more than two populations and informations from more than two samples. The main aim of our design of experiment is to control the extraneous variables & to minimize the experimental error so that the results of our experiments could be attributed to the experimental variables only.

**Treatments :** Various objects of comparison in a comparative experiment are called treatments. **Eg:** In an agricultural experiment, different fertilizers used or different types of crops or different methods of cultivation are the treatments.

**Blocks :** In an agricultural experiment, we divide the whole experimental units or fields, into relatively homogeneous subgroups or strata. These subgroups which are more homogeneous amongst themselves than the whole field.

**Experimental Units:** The smallest division of the experimental material to which we apply the treatments and on which we make observations on the experimental variables under the investigation is called experimental unit. For example, in an agricultural experiment, the plot of land is the experimental unit.

**Experimental Error:** Here the error does not mean a mistake but it is due to extraneous variables. In our above agricultural experiment, even if the same treatment, say, fertilizer is used on all the plots of lands, the yield would vary from plot to plot, due to the difference in soil fertility. Such variation in the yield from plot to plot is due to random causes or chance causes or non assignable factors which are beyond our control. This is termed as experimental error.

**ANALYSIS OF VARIANCE (ANOVA)**

**Anova:** The analysis of variance is a widely used technique developed by Prof. R.A. Fisher. It enables us to separate the variation ascribable to one set of causes from the variation ascribable to other set. In other words, it divides the total variation in a group into parts which are ascribable to different factors and residual random variations which could not be ascribed for by any of these factors. The total variation is split up into the following two components. (i) Variation within the subgroups of samples (ii) Variation between the subgroups of the samples. The variation due to any specific factor is compared with the residual variation for the significance by using F test.

**Uses:** It is used to test whether the means of a number of populations (more than two) are equal.

**Anova Table:** A table showing the source of variation, the sum of squares, degree of freedom, mean square (variance) and the formula for F test ratio is called an anova table.

**Assumption:** (i) Each sample taken is a random sample (ii) Each one is independent of the other sample. (iii) Populations from which samples are taken are normal (iv) Variances of the populations are equal (v) The effect of various components are additive.

**Classification of ANOVA:**

- (i) One way classification (or) One factor of classification (or) One way ANOVA  
(or) Completely Randomized design
- (ii) Two way classification (or) Two factors of classification (or) Manifold classification  
(or) Randomized block design.



## ANALYSIS OF VARIANCE (ANOVA) FOR ONE WAY CLASSIFICATION

### Working Procedure

1.  $H_0$ : There is no significant difference between the treatments.
2.  $H_1$ : There is significant difference between the treatments.
3. Find  $N$ , No. of items in the given data.
4. Find  $T$ , Sum of all the items.
5. Find  $\frac{T^2}{N}$ , Correction factor.
6. Calculate the total sum of squares.  $TSS = \sum x_1^2 + \sum x_2^2 + \sum x_3^2 - \frac{T^2}{N}$
7. Calculate the column sum of squares.  $SSC = \frac{(\sum x_1)^2}{n_1} + \frac{(\sum x_2)^2}{n_2} + \frac{(\sum x_3)^2}{n_3} - \frac{T^2}{N}$   
where  $n_1, n_2, n_3$  is the no. of elements in each column.
8. Calculate the sum of squares within samples.  $SSE = TSS - SSC$
9. Prepare the ANOVA table to calculate F ratio
10. Find the table value
11. Conclusion

1. The following data are per hectare yield for three varieties of wheat, each grown in four plots. Using one way ANOVA table test whether there is a significant difference between the average yields in the 3 varieties of wheat.

| Plots of Land | Variety of wheat |   |   |
|---------------|------------------|---|---|
|               | A                | B | C |
| 1             | 6                | 5 | 5 |
| 2             | 7                | 5 | 4 |
| 3             | 3                | 3 | 3 |
| 4             | 8                | 7 | 4 |

**Solution:**  $H_0$ : There is no significant difference.  $H_1$ : There is significant difference.

| $x_1$           | $x_2$           | $x_3$           | Total    | $x_1^2$            | $x_2^2$            | $x_3^2$           |
|-----------------|-----------------|-----------------|----------|--------------------|--------------------|-------------------|
| 6               | 5               | 5               | 16       | 36                 | 25                 | 25                |
| 7               | 5               | 4               | 16       | 49                 | 25                 | 16                |
| 3               | 3               | 3               | 9        | 9                  | 9                  | 9                 |
| 8               | 7               | 4               | 19       | 64                 | 49                 | 16                |
| $\sum x_1 = 24$ | $\sum x_2 = 20$ | $\sum x_3 = 16$ | $T = 60$ | $\sum x_1^2 = 158$ | $\sum x_2^2 = 108$ | $\sum x_3^2 = 66$ |

$N$  = Total no. of items in the given data = 12,

$T$  = Sum of all the items = 60

$$\frac{T^2}{N} = \text{Correction Factor (C.F.)} = \frac{60^2}{12} = 300$$

$$TSS = \text{Total sum of squares} = \sum x_1^2 + \sum x_2^2 + \sum x_3^2 - \frac{T^2}{N} = 158 + 108 + 66 - 300 = 32$$

$$SSC = \text{Sum of squares between samples} = \frac{(\sum x_1)^2}{n_1} + \frac{(\sum x_2)^2}{n_2} + \frac{(\sum x_3)^2}{n_3} - \frac{T^2}{N} = \frac{24^2}{4} + \frac{20^2}{4} + \frac{16^2}{4} - 300 = 8$$

$$SSE = \text{Sum of squares within samples} = TSS - SSC = 32 - 8 = 24$$

$$C = \text{No. of samples} = 3, C - 1 = 3 - 1 = 2, N - C = 12 - 3 = 9$$

$$MSC = \text{Mean of Squares between samples} = \frac{SSC}{C-1} = \frac{8}{2} = 4$$

$$MSE = \text{Mean of Squares within samples} = \frac{SSE}{N-C} = \frac{24}{9} = 2.66$$

### ANOVA Table

| Source of Variation | Sum of Squares | Degree of Freedom | Mean Sum of Squares | Variance Ratio                | Table Value at 5% Level of Significance |
|---------------------|----------------|-------------------|---------------------|-------------------------------|-----------------------------------------|
| Between Treatments  | $SSC = 8$      | $C - 1 = 2$       | $MSC = 4$           | $F = \frac{MSC}{MSE} = 1.503$ | $F(2, 9) = 4.26$                        |
| Error               | $SSE = 24$     | $N - C = 9$       | $MSE = 2.66$        |                               |                                         |

Calculated  $F < \text{Tabulated } F$ ,  $H_0$  accepted.

### ANALYSIS OF VARIANCE (ANOVA) FOR TWO WAY CLASSIFICATION

#### Working Procedure

1.  $H_0$ : There is no significant difference between the treatments.
2.  $H_1$ : There is significant difference between the treatments.
3. Find  $N$ , No. of items in the given data. 4. Find  $T$ , Sum of all the items. 5. Find  $\frac{T^2}{N}$ , Correction factor.
6. Calculate the total sum of squares.  $TSS = \sum x_1^2 + \sum x_2^2 + \sum x_3^2 - \frac{T^2}{N}$
7. Calculate the column sum of squares.  $SSC = \frac{(\sum x_1)^2}{n_1} + \frac{(\sum x_2)^2}{n_2} + \frac{(\sum x_3)^2}{n_3} - \frac{T^2}{N}$ ,  $n_1$  - no. of elements in each column.
8. Calculate the row sum of squares.  $SSR = \frac{(\sum y_1)^2}{n_2} + \frac{(\sum y_2)^2}{n_2} + \frac{(\sum y_3)^2}{n_2} - \frac{T^2}{N}$ , where  $n_2$  - no. of elements in each row.
9. Calculate the sum of squares within samples.  $SSE = TSS - SSC - SSR$
10. Prepare the ANOVA table to calculate  $F$  ratio.
11. Find the table value  $F_C$  and  $F_R$ .
12. Conclusion

### PROBLEMS IN TWO WAY CLASSIFICATION

1. Three varieties of a crop are tested in a randomized block design with four replications, the layout being as given below: The yields are given in kilograms. Analyze for significance.

| Blocks | A  | B  | C  |
|--------|----|----|----|
| 1      | 47 | 49 | 48 |
| 2      | 51 | 49 | 53 |
| 3      | 49 | 52 | 52 |
| 4      | 49 | 50 | 51 |

**Solution:**

| Blocks | A  | B  | C  |
|--------|----|----|----|
| 1      | 47 | 49 | 48 |
| 2      | 51 | 49 | 53 |
| 3      | 49 | 52 | 52 |
| 4      | 49 | 50 | 51 |

To make the computation simpler, subtract 50 from each given value.

| Blocks | A  | B  | C  |
|--------|----|----|----|
| 1      | -3 | -1 | -2 |
| 2      | 1  | -1 | 3  |
| 3      | -1 | 2  | 2  |
| 4      | -1 | 0  | 1  |

$H_0$ : There is no significant difference.  $H_1$ : There is significant difference.

|       | $x_1$           | $x_2$          | $x_3$          | Total           | $x_1^2$           | $x_2^2$          | $x_3^2$           |
|-------|-----------------|----------------|----------------|-----------------|-------------------|------------------|-------------------|
| $y_1$ | -3              | -1             | -2             | $\sum y_1 = -6$ | 9                 | 1                | 4                 |
| $y_2$ | 1               | -1             | 3              | $\sum y_2 = 3$  | 1                 | 1                | 9                 |
| $y_3$ | -1              | 2              | 2              | $\sum y_3 = 3$  | 1                 | 4                | 4                 |
| $y_4$ | -1              | 0              | 1              | $\sum y_4 = 0$  | 1                 | 0                | 1                 |
|       | $\sum x_1 = -4$ | $\sum x_2 = 0$ | $\sum x_3 = 4$ | $T = 0$         | $\sum x_1^2 = 12$ | $\sum x_2^2 = 6$ | $\sum x_3^2 = 18$ |

$N$  = Total no. of items in the given data = 12,  $T$  = Sum of all the items = 0

$$\frac{T^2}{N} = \text{Correction Factor (C.F.)} = \frac{0^2}{12} = 0$$

$$TSS = \text{Total sum of squares} = \sum x_1^2 + \sum x_2^2 + \sum x_3^2 - \frac{T^2}{N} = 12 + 6 + 18 - 0 = 36$$

$$SSC = \text{Sum of squares between column} = \frac{(\sum x_1)^2}{n_1} + \frac{(\sum x_2)^2}{n_1} + \frac{(\sum x_3)^2}{n_1} - \frac{T^2}{N} = \frac{(-4)^2}{4} + \frac{(0)^2}{4} + \frac{(4)^2}{4} - 0 = 8$$

$SSR$  = Sum of squares between rows

$$SSR = \frac{(\sum y_1)^2}{n_2} + \frac{(\sum y_2)^2}{n_2} + \frac{(\sum y_3)^2}{n_2} + \frac{(\sum y_4)^2}{n_2} - \frac{T^2}{N} = \frac{(-6)^2}{3} + \frac{(3)^2}{3} + \frac{(3)^2}{3} + \frac{(0)^2}{3} - 0 = 18$$

$$SSE = \text{Sum of squares within samples} = TSS - SSC - SSR = 36 - 8 - 18 = 10$$

$c$  = No. of columns,  $r$  = No. of rows,  $r - 1 = 4 - 1 = 3$ ,  $c - 1 = 3 - 1 = 2$

$$MSC = \text{Mean of Squares between column} = \frac{SSC}{c-1} = \frac{8}{2} = 4$$

$$MSR = \text{Mean of Squares between rows} = \frac{SSR}{r-1} = \frac{18}{3} = 6$$

$$MSE = \text{Mean of Squares within samples} = \frac{SSE}{(c-1)(r-1)} = \frac{10}{6} = 1.67$$

#### ANOVA Table

| Source of Variation     | Sum of Squares | Degree of Freedom    | Mean Sum of Squares | Variance Ratio                | Table Value at 5% Level of Significance |
|-------------------------|----------------|----------------------|---------------------|-------------------------------|-----------------------------------------|
| Between Columns (Crops) | $SSC = 8$      | $c - 1 = 2$          | $MSC = 4$           | $F_C = \frac{MSC}{MSE} = 2.4$ | $F(2, 6) = 5.14$                        |
| Between Rows (Blocks)   | $SSR = 18$     | $r - 1 = 3$          | $MSR = 6$           |                               |                                         |
| Residual                | $SSE = 10$     | $(c - 1)(r - 1) = 6$ | $MSE = 1.67$        | $F_R = \frac{MSR}{MSE} = 3.6$ | $F(3, 6) = 4.76$                        |

(i) Calculated  $F_C < \text{Tabulated } F$ ,  $H_0$  accepted.

(ii) Calculated  $F_R < \text{Tabulated } F$ ,  $H_0$  accepted.

2. Four doctors each test four treatments for a certain disease and observe the number of days each patient takes to recover. The results are as follows (recovery time in days). Discuss the difference between doctors and treatments

| Doctor | Treatment |    |    |    |
|--------|-----------|----|----|----|
|        | 1         | 2  | 3  | 4  |
| A      | 10        | 14 | 19 | 20 |
| B      | 11        | 15 | 17 | 21 |
| C      | 9         | 12 | 16 | 19 |
| D      | 8         | 13 | 17 | 20 |

**Solution:** To make the computation simpler, subtract 15 from each given value.

| Doctor | Treatment |    |   |   |
|--------|-----------|----|---|---|
|        | 1         | 2  | 3 | 4 |
| A      | -5        | -1 | 4 | 5 |
| B      | -4        | 0  | 2 | 6 |
| C      | -6        | -3 | 1 | 4 |
| D      | -7        | -2 | 2 | 5 |

$H_0$ : There is no significant difference.  $H_1$ : There is significant difference.

|       | $x_1$            | $x_2$           | $x_3$          | $x_4$           | Total           | $x_1^2$            | $x_2^2$           | $x_3^2$           | $x_4^2$            |
|-------|------------------|-----------------|----------------|-----------------|-----------------|--------------------|-------------------|-------------------|--------------------|
| $y_1$ | -5               | -1              | 4              | 5               | $\sum y_1 = 3$  | 25                 | 1                 | 16                | 25                 |
| $y_2$ | -4               | 0               | 2              | 6               | $\sum y_2 = 4$  | 16                 | 0                 | 4                 | 36                 |
| $y_3$ | -6               | -3              | 1              | 4               | $\sum y_3 = -4$ | 36                 | 9                 | 1                 | 16                 |
| $y_4$ | -7               | -2              | 2              | 5               | $\sum y_4 = -2$ | 49                 | 4                 | 4                 | 25                 |
|       | $\sum x_1 = -22$ | $\sum x_2 = -6$ | $\sum x_3 = 9$ | $\sum x_4 = 20$ | $T = 1$         | $\sum x_1^2 = 126$ | $\sum x_2^2 = 14$ | $\sum x_3^2 = 25$ | $\sum x_4^2 = 102$ |

$N$  = Total no. of items in the given data = 16,  $T$  = Sum of all the items = 1

$$\frac{T^2}{N} = \text{Correction Factor (C.F.)} = \frac{1^2}{16} = 0.0625$$

$$TSS = \text{Total sum of squares} = \sum x_1^2 + \sum x_2^2 + \sum x_3^2 + \sum x_4^2 - \frac{T^2}{N} = 126 + 14 + 25 + 102 - 0.0625 = 266.94$$

$$SSC = \frac{(\sum x_1)^2}{n_1} + \frac{(\sum x_2)^2}{n_1} + \frac{(\sum x_3)^2}{n_1} + \frac{(\sum x_4)^2}{n_1} - \frac{T^2}{N} = \frac{(-22)^2}{4} + \frac{(-6)^2}{4} + \frac{(9)^2}{4} + \frac{(20)^2}{4} - 0.0625 = 250.19$$

$$SSR = \frac{(\sum y_1)^2}{n_2} + \frac{(\sum y_2)^2}{n_2} + \frac{(\sum y_3)^2}{n_2} + \frac{(\sum y_4)^2}{n_2} - \frac{T^2}{N} = \frac{(3)^2}{4} + \frac{(4)^2}{4} + \frac{(-4)^2}{4} + \frac{(-2)^2}{4} - 0.0625 = 11.19$$

$$SSE = \text{Sum of squares within samples} = TSS - SSC - SSR = 5.56$$

$$c = \text{No. of columns}, r = \text{No. of rows}, r - 1 = 4 - 1 = 3, c - 1 = 4 - 1 = 3$$

$$MSC = \text{Mean of Squares between column} = \frac{SSC}{c-1} = \frac{250.19}{3} = 83.4$$

$$MSR = \text{Mean of Squares between rows} = \frac{SSR}{r-1} = \frac{11.19}{3} = 3.73$$

$$MSE = \text{Mean of Squares within samples} = \frac{SSE}{(c-1)(r-1)} = \frac{5.56}{9} = 0.62$$

#### ANOVA Table

| Source of Variation     | Sum of Squares | Degree of Freedom    | Mean Sum of Squares | Variance Ratio                                                         | Table Value at 5% |
|-------------------------|----------------|----------------------|---------------------|------------------------------------------------------------------------|-------------------|
| Between Columns (Crops) | $SSC = 250.19$ | $c - 1 = 3$          | $MSC = 83.4$        | $F_C = \frac{MSC}{MSE} = 134.52$<br><br>$F_R = \frac{MSR}{MSE} = 6.02$ | $F(3, 9) = 3.86$  |
| Between Rows (Blocks)   | $SSR = 11.19$  | $r - 1 = 3$          | $MSR = 3.73$        |                                                                        |                   |
| Residual                | $SSE = 5.56$   | $(c - 1)(r - 1) = 9$ | $MSE = 0.62$        |                                                                        |                   |

(i) Calculated  $F_C < \text{Tabulated } F$ ,  $H_0$  rejected. (ii) Calculated  $F_R < \text{Tabulated } F$ ,  $H_0$  rejected.

**t - test table**

Values of  $|t|$  with probability  $P$  and degrees of freedom  $\nu$

| $P \backslash \nu$ | 0.50  | 0.10 | 0.05  | 0.02  | 0.01  |
|--------------------|-------|------|-------|-------|-------|
| 1                  | 1.000 | 6.34 | 12.71 | 31.82 | 63.66 |
| 2                  | 0.816 | 2.92 | 4.30  | 6.96  | 9.92  |
| 3                  | 0.765 | 2.35 | 3.18  | 4.54  | 5.84  |
| 4                  | 0.741 | 2.13 | 2.78  | 3.75  | 4.60  |
| 5                  | 0.727 | 2.02 | 2.57  | 3.36  | 4.03  |
| 6                  | 0.718 | 1.94 | 2.45  | 3.14  | 3.71  |
| 7                  | 0.711 | 1.90 | 2.36  | 3.00  | 3.50  |
| 8                  | 0.706 | 1.86 | 2.31  | 2.90  | 3.36  |
| 9                  | 0.703 | 1.83 | 2.26  | 2.82  | 3.25  |
| 10                 | 0.700 | 1.81 | 2.23  | 2.76  | 3.17  |
| 11                 | 0.697 | 1.80 | 2.20  | 2.72  | 3.11  |
| 12                 | 0.695 | 1.78 | 2.18  | 2.68  | 3.06  |
| 13                 | 0.694 | 1.77 | 2.16  | 2.65  | 3.01  |
| 14                 | 0.692 | 1.76 | 2.14  | 2.62  | 2.98  |
| 15                 | 0.691 | 1.75 | 2.13  | 2.60  | 2.95  |
| 16                 | 0.690 | 1.75 | 2.12  | 2.58  | 2.92  |
| 17                 | 0.689 | 1.74 | 2.11  | 2.57  | 2.90  |
| 18                 | 0.688 | 1.73 | 2.10  | 2.55  | 2.88  |
| 19                 | 0.688 | 1.73 | 2.09  | 2.54  | 2.86  |
| 20                 | 0.687 | 1.72 | 2.09  | 2.53  | 2.84  |
| 21                 | 0.686 | 1.72 | 2.08  | 2.52  | 2.83  |
| 22                 | 0.686 | 1.72 | 2.07  | 2.51  | 2.82  |
| 23                 | 0.685 | 1.71 | 2.07  | 2.50  | 2.81  |
| 24                 | 0.685 | 1.71 | 2.06  | 2.49  | 2.80  |
| 25                 | 0.684 | 1.71 | 2.06  | 2.48  | 2.79  |
| 26                 | 0.684 | 1.71 | 2.06  | 2.48  | 2.78  |
| 27                 | 0.684 | 1.70 | 2.05  | 2.47  | 2.77  |
| 28                 | 0.683 | 1.70 | 2.05  | 2.47  | 2.76  |
| 29                 | 0.683 | 1.70 | 2.04  | 2.46  | 2.76  |
| 30                 | 0.683 | 1.70 | 2.04  | 2.46  | 2.75  |



# F test Table

5% and 1% points of F.

| v <sub>1</sub> \ v <sub>2</sub> | 1     | 2     | 3     | 4     | 5     | 6     | 8     | 12    | 24    | ∞     |
|---------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 2                               | 18.51 | 19.00 | 19.16 | 19.25 | 19.30 | 19.32 | 19.37 | 19.41 | 19.45 | 19.50 |
|                                 | 98.49 | 99.00 | 99.17 | 99.25 | 99.30 | 99.33 | 99.36 | 99.42 | 99.46 | 99.50 |
| 3                               | 10.13 | 9.55  | 9.28  | 9.12  | 9.01  | 8.94  | 8.84  | 8.74  | 8.64  | 8.53  |
|                                 | 34.12 | 30.82 | 29.46 | 28.71 | 28.24 | 27.91 | 27.49 | 27.05 | 26.60 | 26.12 |
| 4                               | 7.71  | 6.94  | 6.59  | 6.39  | 6.26  | 6.16  | 6.04  | 5.91  | 5.77  | 5.63  |
|                                 | 21.20 | 18.00 | 16.69 | 15.98 | 15.52 | 15.21 | 14.80 | 14.37 | 13.93 | 13.46 |
| 5                               | 6.61  | 5.79  | 5.41  | 5.19  | 5.05  | 4.95  | 4.82  | 4.68  | 4.53  | 4.36  |
|                                 | 16.26 | 13.27 | 12.06 | 11.39 | 10.97 | 10.67 | 10.27 | 9.89  | 9.47  | 9.02  |
| 6                               | 5.99  | 5.14  | 4.76  | 4.53  | 4.39  | 4.28  | 4.15  | 4.00  | 3.84  | 3.67  |
|                                 | 13.74 | 10.92 | 9.78  | 9.15  | 8.75  | 8.47  | 8.10  | 7.72  | 7.31  | 6.88  |
| 7                               | 5.59  | 4.74  | 4.35  | 4.12  | 3.97  | 3.87  | 3.73  | 3.57  | 3.41  | 3.23  |
|                                 | 12.25 | 9.55  | 8.45  | 7.85  | 7.46  | 7.19  | 6.84  | 6.47  | 6.07  | 5.65  |
| 8                               | 5.32  | 4.46  | 4.07  | 3.84  | 3.69  | 3.58  | 3.44  | 3.28  | 3.12  | 2.93  |
|                                 | 11.26 | 8.65  | 7.59  | 7.01  | 6.63  | 6.37  | 6.03  | 5.67  | 5.28  | 4.86  |
| 9                               | 5.12  | 4.26  | 3.86  | 3.63  | 3.48  | 3.37  | 3.23  | 3.07  | 2.90  | 2.71  |
|                                 | 10.56 | 8.02  | 6.99  | 6.42  | 6.06  | 5.80  | 5.47  | 5.11  | 4.73  | 4.31  |
| 10                              | 4.96  | 4.10  | 3.71  | 3.48  | 3.33  | 3.22  | 3.07  | 2.91  | 2.74  | 2.54  |
|                                 | 10.04 | 7.56  | 6.55  | 5.99  | 5.64  | 5.39  | 5.06  | 4.71  | 4.33  | 3.91  |
| 12                              | 4.75  | 3.88  | 3.49  | 3.26  | 3.11  | 3.00  | 2.85  | 2.69  | 2.50  | 2.30  |
|                                 | 9.33  | 6.93  | 5.95  | 5.41  | 5.06  | 4.82  | 4.50  | 4.16  | 3.78  | 3.36  |
| 14                              | 4.60  | 3.74  | 3.34  | 3.11  | 2.96  | 2.85  | 2.70  | 2.53  | 2.35  | 2.13  |
|                                 | 8.86  | 6.51  | 5.56  | 5.03  | 4.69  | 4.46  | 4.14  | 3.80  | 3.43  | 3.00  |
| 16                              | 4.49  | 3.63  | 3.24  | 3.01  | 2.85  | 2.74  | 2.59  | 2.42  | 2.24  | 2.01  |
|                                 | 8.53  | 6.23  | 5.29  | 4.77  | 4.44  | 4.20  | 3.89  | 3.55  | 3.18  | 2.75  |
| 18                              | 4.41  | 3.55  | 3.16  | 2.93  | 2.77  | 2.66  | 2.51  | 2.34  | 2.15  | 1.92  |
|                                 | 8.28  | 6.01  | 5.09  | 4.58  | 4.25  | 4.01  | 3.71  | 3.37  | 3.01  | 2.57  |
| 20                              | 4.35  | 3.49  | 3.10  | 2.87  | 2.71  | 2.60  | 2.45  | 2.28  | 2.08  | 1.84  |
|                                 | 8.10  | 5.85  | 4.94  | 4.43  | 4.10  | 3.87  | 3.56  | 3.23  | 2.86  | 2.42  |
| 25                              | 4.24  | 3.38  | 2.99  | 2.76  | 2.60  | 2.49  | 2.34  | 2.16  | 1.96  | 1.71  |
|                                 | 7.77  | 5.57  | 4.68  | 4.18  | 3.86  | 3.63  | 3.32  | 2.99  | 2.62  | 2.17  |
| 30                              | 4.17  | 3.32  | 2.92  | 2.69  | 2.53  | 2.42  | 2.27  | 2.09  | 1.89  | 1.62  |
|                                 | 7.56  | 5.39  | 4.51  | 4.02  | 3.70  | 3.47  | 3.17  | 2.84  | 2.47  | 2.01  |
| 40                              | 4.08  | 3.23  | 2.84  | 2.61  | 2.45  | 2.34  | 2.18  | 2.00  | 1.79  | 1.51  |
|                                 | 7.31  | 5.18  | 4.31  | 3.83  | 3.51  | 3.29  | 2.99  | 2.66  | 2.29  | 1.81  |
| 60                              | 4.00  | 3.15  | 2.76  | 2.52  | 2.37  | 2.25  | 2.10  | 1.92  | 1.70  | 1.39  |
|                                 | 7.08  | 4.98  | 4.13  | 3.65  | 3.34  | 3.12  | 2.82  | 2.50  | 2.12  | 1.60  |

# Chi-Square Test Table

Values of  $\chi^2$  with probability  $P$  and df  $v$

| $P \backslash v$ | 0.99   | 0.95  | 0.50  | 0.30  | 0.20  | 0.10  | 0.05  | 0.01  |
|------------------|--------|-------|-------|-------|-------|-------|-------|-------|
| 1                | 0.0002 | 0.004 | 0.46  | 1.07  | 1.64  | 2.71  | 3.84  | 6.64  |
| 2                | 0.020  | 0.103 | 1.39  | 2.41  | 3.22  | 4.60  | 5.99  | 9.21  |
| 3                | 0.115  | 0.35  | 2.37  | 3.66  | 4.64  | 6.25  | 7.82  | 11.34 |
| 4                | 0.30   | 0.71  | 3.36  | 4.88  | 5.99  | 7.78  | 9.49  | 13.28 |
| 5                | 0.55   | 1.14  | 4.35  | 6.06  | 7.29  | 9.24  | 11.07 | 15.09 |
| 6                | 0.87   | 1.64  | 5.35  | 7.23  | 8.56  | 10.64 | 12.59 | 16.81 |
| 7                | 1.24   | 2.17  | 6.35  | 8.38  | 9.80  | 12.02 | 14.07 | 18.48 |
| 8                | 1.65   | 2.73  | 7.34  | 9.52  | 11.03 | 13.36 | 15.51 | 20.09 |
| 9                | 2.09   | 3.32  | 8.34  | 10.66 | 12.24 | 14.68 | 16.92 | 21.67 |
| 10               | 2.56   | 3.94  | 9.34  | 11.78 | 13.44 | 15.99 | 18.31 | 23.21 |
| 11               | 3.05   | 4.58  | 10.34 | 12.90 | 14.63 | 17.28 | 19.68 | 24.72 |
| 12               | 3.57   | 5.23  | 11.34 | 14.01 | 15.81 | 18.55 | 21.03 | 26.22 |
| 13               | 4.11   | 5.89  | 12.34 | 15.12 | 16.98 | 19.81 | 22.36 | 27.69 |
| 14               | 4.66   | 6.57  | 13.34 | 16.22 | 18.15 | 21.06 | 23.68 | 29.14 |
| 15               | 5.23   | 7.26  | 14.34 | 17.32 | 19.31 | 22.31 | 25.00 | 30.58 |
| 16               | 5.81   | 7.96  | 15.34 | 18.42 | 20.46 | 23.54 | 26.30 | 32.00 |
| 17               | 6.41   | 8.67  | 16.34 | 19.51 | 21.62 | 24.77 | 27.59 | 33.41 |
| 18               | 7.02   | 9.39  | 17.34 | 20.60 | 22.76 | 25.99 | 28.87 | 34.80 |
| 19               | 7.63   | 10.12 | 18.34 | 21.69 | 23.90 | 27.20 | 30.14 | 36.19 |
| 20               | 8.26   | 10.85 | 19.34 | 22.78 | 25.04 | 28.41 | 31.41 | 37.57 |
| 21               | 8.90   | 11.59 | 20.34 | 23.86 | 26.17 | 29.62 | 32.67 | 38.93 |
| 22               | 9.54   | 12.34 | 21.34 | 24.94 | 27.30 | 30.81 | 33.92 | 40.29 |
| 23               | 10.20  | 13.09 | 22.34 | 26.02 | 28.43 | 32.01 | 35.17 | 41.64 |
| 24               | 10.86  | 13.85 | 23.34 | 27.10 | 29.55 | 33.20 | 36.42 | 42.98 |
| 25               | 11.52  | 14.61 | 24.34 | 28.17 | 30.68 | 34.68 | 37.65 | 44.31 |
| 26               | 12.20  | 15.38 | 25.34 | 29.25 | 31.80 | 35.56 | 38.88 | 45.64 |
| 27               | 12.88  | 16.15 | 26.34 | 30.32 | 32.91 | 36.74 | 40.11 | 46.96 |
| 28               | 13.56  | 16.93 | 27.34 | 31.39 | 34.03 | 37.92 | 41.34 | 48.28 |
| 29               | 14.26  | 17.71 | 28.34 | 32.46 | 35.14 | 39.09 | 42.56 | 49.59 |
| 30               | 14.95  | 18.49 | 29.34 | 33.53 | 36.25 | 40.26 | 43.77 | 50.89 |